Titration –I

(Link $KMnO_4$)Vs (Standard FAS)

S.no	Volume of standard FAS (ml)	Burette readings		Concordant value (Volume of $KMnO_4$) (ml)
		Initial (ml)	Final (ml)	
1	20			
2	20			
3	20			

Calculation :

Volume of $KMnO_4$ (link) solution (V_1) = ----------ml

Normality $KMnO_4$ (link) solution (N_1) =----------N

Volume of standard FAS solution (V_2) = 20 ml

Normality of standard FAS solution (N_2) = 0.1102 N

According to normality equation: $V_1 \times N_1 = V_2 \times N_2$

$$N_1 = \frac{V_2 \times N_2}{V_1}$$

Normality of $KMnO_4$ (link) solution (N_1) = _____X_____ N

Titration–II

(Unknown $FeSO_4$) Vs (Link $KMnO_4$)

Burette is washed with water, rinsed with $KMnO_4$ solution and filled with same $KMnO_4$ solution up to the zero mark. Exactly 20 ml of unknown $FeSO_4$ solution is pipetted out into the clean, washed conical flask. To this $FeSO_4$ solution approximately 20ml of 2N sulphuric acid is added. This mixture is titrated against $KMnO_4$ Link solution from the burette. $KMnO_4$ is added drop wise till the appearance of permanent pale pink colour. Burette reading is noted and the same procedure is repeated to get concordant values.

organic analysis.indd 266 19-01-2021 12:24:22